

Independent Mathematica Derivation for $D_{s1}/B_{s1} \rightarrow D_s/B_s \gamma$

Ds/Bs gamma project

24 July 2026

Abstract

This note is a clean, independent restart of the symbolic part of the $D_{s1}/B_{s1} \rightarrow D_s/B_s \gamma$ light-cone sum-rule calculation. It records exactly what has been implemented and checked in *Mathematica* and FEYN CALC: the currents, correlator ordering, Wick contraction, propagator routing, Rohrwild three-point/background-field notation, hard-emission traces, Ward identities, E1 projection, inverse-denominator reduction, Feynman parameterization, the first soft two-particle Fierz traces, physical-state mixing, the pure-axial Colangelo limit, the decay-constant normalization, the complete axial and tensor post-double-Borel invariants, and the numerical couplings and widths. The Mathematica numerical implementation is independent of the Python implementation; their output tables are compared only after both calculations have finished. The 24–26 July revision separates off-shell QCD kinematics from on-shell pole kinematics and withdraws the earlier physical-mass substitution in the theoretical tensor numerator.

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1 Purpose and executable files

The interactive source of this calculation is

`notebooks/DsBs_gamma_symbolic_derivation.nb.`

It contains 53 input cells designed to be evaluated from top to bottom or one at a time. The noninteractive regression scripts are

`scripts/mathematica_current_wick_trace_audit.wl,`
`scripts/mathematica_correlator_routing_audit.wl,`
`scripts/mathematica_hard_loop_parameterization.wl,`
`scripts/mathematica_state_mixing_projection.wl,`
`scripts/step14_explicit_double_borel_forms.wl,`
`scripts/step15_complete_three_particle_borel.wl.`

The first three batch audits write their full machine-readable expressions under `outputs/`; the state-mixing audit prints its exact residuals and interval overlaps directly. This note presents the compact analytic content needed to understand and reproduce those expressions.

2 Conventions and kinematics

We use

$$g_{\mu\nu} = \text{diag}(1, -1, -1, -1), \quad \{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}, \quad (1)$$

$$\sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu], \quad \epsilon^{0123} = +1. \quad (2)$$

The FEYNCalc convention is explicitly checked through `FeynCalc['LeviCivitaSign=-1]`.

The momentum assignment is

$$p' = p + q, \quad (3)$$

where p' is the initial 1^+ momentum, p is the final 0^- momentum, and q is the photon momentum. The photon is real,

$$q^2 = 0, \quad 2p \cdot q = p'^2 - p^2. \quad (4)$$

The heavy flavor is denoted by $Q = c, b$ and its mass by m_Q ; the light flavor is strange with mass m_s .

3 Currents and the Rohrwild correlator

The two basis-current vertices are

$$\Gamma_\mu^A = \gamma_\mu \gamma_5, \quad (5)$$

$$\Gamma_\mu^B = \frac{i}{m_Q + m_s} \sigma_{\mu\alpha} p'^\alpha \gamma_5, \quad (6)$$

and the pseudoscalar vertex is

$$\Gamma_P = i\gamma_5. \quad (7)$$

The tensor vertex is independently represented as

$$\Gamma_\mu^B = -\frac{(\gamma_\mu \not{p}' - \not{p}' \gamma_\mu) \gamma_5}{2(m_Q + m_s)}. \quad (8)$$

Mathematica verifies that the two definitions agree exactly.

The physical-current rotations are kept outside the basis traces:

$$\Gamma_\mu^{\text{low}} = \sin \theta \Gamma_\mu^A + \cos \theta \Gamma_\mu^B, \quad (9)$$

$$\Gamma_\mu^{\text{high}} = \cos \theta \Gamma_\mu^A - \sin \theta \Gamma_\mu^B. \quad (10)$$

Following Rohrwild's vacuum three-point notation [arXiv:0708.1405](#), adapted to a $1^+ \rightarrow 0^- \gamma$ transition, the starting correlator is

$$\Pi_{\mu\nu}^X(p, q) = i^2 \int d^4x d^4y e^{ip \cdot x + iq \cdot y} \langle 0 | T \{ J_P^\dagger(x) j_\nu^{\text{em}}(y) J_\mu^X(0) \} | 0 \rangle, \quad X = A, B, \quad (11)$$

where

$$j_\nu^{\text{em}} = e_s \bar{s} \gamma_\nu s + e_Q \bar{Q} \gamma_\nu Q. \quad (12)$$

Here e_s, e_Q are charges in units of the positive elementary charge e . Introduce the plane-wave background field

$$F_{\alpha\beta}(z) = i(q_\alpha \varepsilon_\beta^* - q_\beta \varepsilon_\alpha^*) e^{iq \cdot z}. \quad (13)$$

The term linear in this field obeys

$$\varepsilon^{*\nu} \Pi_{\mu\nu}^X(p, q) = i \int d^4x e^{ip \cdot x} \langle 0 | T \{ J_P^\dagger(x) J_\mu^X(0) \} | 0 \rangle_F \equiv i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ J_P^\dagger(x) J_\mu^X(0) \} | 0 \rangle, \quad (14)$$

up to the explicitly stated overall QED convention. Thus the former vacuum-to-photon formula was not a different calculation; it is the single-photon component of Eq. (11).

In explicit fields the ordered product is

$$T \{ \bar{s}(x) \Gamma_P Q(x) \bar{Q}(0) \Gamma_\mu^X s(0) \}. \quad (15)$$

Thus the vertex at x carries the final channel p , while the current at the origin carries the initial channel $p' = p + q$.

The current definitions used in the notebook are:

```

gammaA[index_] := GA[index] . GA[5];
gammaB[index_] :=
  I FV[pPrime, alpha] DiracSigma[GA[index], GA[alpha]] . GA[5]/
  (mQ + ms);
gammaBExpanded[index_] :=
  -(GA[index] . GS[pPrime] - GS[pPrime] . GA[index]) . GA[5]/
  (2 (mQ + ms));
gammaP[] := I GA[5];
propNumerator[momentum_, mass_] := GS[momentum] + mass;

```

4 Wick contraction and global signs

For the complete hard contraction, Eq. (15) gives

$$\langle T\{\bar{s}\Gamma_P Q \bar{Q}\Gamma_X s\}\rangle = -\text{Tr}_D[\Gamma_P S_Q \Gamma_X S_s]. \quad (16)$$

The minus sign is the closed-fermion-loop sign. Including the leading correlator factor i , but not yet the propagator or QED-vertex factors, the hard prefactor is therefore

$$-iN_c. \quad (17)$$

If only the heavy flavor is contracted, the strange bilocal remains open and there is no closed fermion loop. The soft contraction begins with $+i$. The Fierz rearrangement

$$\bar{s}_\alpha(x) s_\beta(0) = -\frac{1}{4} \sum_i (\Gamma_i)_{\beta\alpha} \bar{s}(x) \Gamma_i s(0) \quad (18)$$

then gives a soft prefactor $-i/4$ before the free-propagator factor.

In the notebook the bookkeeping is kept visible:

```

fermionLoopSign = -1;
hardWickPrefactor = I fermionLoopSign Nc;
softWickPrefactorBeforeFierz = I;
softWickPrefactorAfterFierz = -I/4;

```

5 Propagator expansion and OPE building blocks

The free momentum-space propagator is

$$S_f^0(r) = \frac{i(\not{r} + m_f)}{r^2 - m_f^2 + i0}. \quad (19)$$

5.1 Light-strange propagator in Rohrwild notation, including m_s

To make the comparison with Rohrwild direct, write $\bar{u} = 1 - u$ and keep the background fields at the point ux . With color indices displayed, the light-strange propagator used as the master

expansion is

$$\begin{aligned}
s^a \widehat{(x) \bar{s}^b}(0) &= \delta^{ab} \left[\frac{i \not{x}}{2\pi^2 x^4} - \frac{m_s}{4\pi^2 x^2} \right] \\
&- \frac{ig_s}{16\pi^2 x^2} \int_0^1 du \{ \bar{u} \not{x} \sigma_{\alpha\beta} + u \sigma_{\alpha\beta} \not{x} \} G_A^{\alpha\beta}(ux) t_{ab}^A \\
&- \frac{ie_s \delta^{ab}}{16\pi^2 x^2} \int_0^1 du \{ \bar{u} \not{x} \sigma_{\alpha\beta} + u \sigma_{\alpha\beta} \not{x} \} F^{\alpha\beta}(ux) \\
&- \frac{m_s}{32\pi^2} \int_0^1 du \left[g_s G_A^{\alpha\beta}(ux) t_{ab}^A + e_s \delta^{ab} F^{\alpha\beta}(ux) \right] \sigma_{\alpha\beta} L_x + \dots, \quad (20)
\end{aligned}$$

where

$$L_x = \ln \left(\frac{-x^2 \mu^2}{4} \right) + 2\gamma_E. \quad (21)$$

The first line contains only the free propagator and its explicit m_s term. The last line is the linear- m_s Wilson coefficient in a single gluon or electromagnetic background field. No local vacuum-condensate term is written in Eq. (20). Setting m_s and the last line to zero gives precisely the light-propagator form displayed in Eq. (2.12) of Rohrwild. The ellipsis denotes higher-background-field and higher-order mass terms, not local condensate pieces used elsewhere.

Important calculation warning. Although local parts are no longer displayed as part of the propagator, the existing axial numerical sum-rule implementation *does include* the term labeled `heavy_local` in the Python audit. After the double Borel transform it is

$$\Pi_{Q\gamma}^{A,\text{local}} = e_Q e^{-m_Q^2/M^2} \langle \bar{s}s \rangle \left[1 - \frac{m_Q m_s}{M^2} + \frac{m_s^2}{2M^2} \left(1 + \frac{m_Q^2}{M^2} \right) \right]. \quad (22)$$

It has not been removed here, because doing so would change the quoted form factors and widths. The mixed condensate $\langle \bar{s} g_s \sigma \cdot G s \rangle$ and the vacuum gluon condensate were not included in the radiative three-point calculation.

5.2 Heavy propagator in a background gluon field

For reference, the momentum representation through one background gluon can be organized as

$$S_Q^{ab}(x) = i \int \frac{d^4 k}{(2\pi)^4} e^{-ik \cdot x} \left\{ \frac{\delta^{ab}(\not{k} + m_Q)}{k^2 - m_Q^2} - \frac{g_s G_{\alpha\beta}^A t_{ab}^A}{4} \frac{\sigma^{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q) \sigma^{\alpha\beta}}{(k^2 - m_Q^2)^2} + \dots \right\}. \quad (23)$$

For the photon-DA calculation the one-gluon term is used as an operator insertion, rather than replaced by $\langle G^2 \rangle$. The explicit light-cone form used in the symbolic three-particle kernels is

$$\begin{aligned}
S_Q^G(x) &= -ig_s \int \frac{d^4 k}{(2\pi)^4} e^{-ik \cdot x} \int_0^1 du \left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2} G^{\alpha\beta}(ux) \sigma_{\alpha\beta} \right. \\
&\quad \left. + \frac{ux_\alpha}{m_Q^2 - k^2} G^{\alpha\beta}(ux) \gamma_\beta \right]. \quad (24)
\end{aligned}$$

Both the $\sigma_{\alpha\beta}$ part and the $x_\alpha \gamma_\beta$ part have Mathematica trace-kernel scripts.

5.3 Electromagnetic background-field propagator

The first-order electromagnetic correction is

$$S_f^\gamma(x) = -ie_f \int \frac{d^4 k}{(2\pi)^4} e^{-ik \cdot x} \int_0^1 du \left[\frac{k + m_f}{2(m_f^2 - k^2)^2} F^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha}{m_f^2 - k^2} F^{\alpha\beta}(ux) \gamma_\beta \right]. \quad (25)$$

The overall electromagnetic coupling may either be included in e_f or kept outside the propagator, but not both. In the explicit-vertex trace audit, Eq. (25) is represented equivalently by

$$S_f^0(r)(-ie_f \gamma_\nu \varepsilon^\nu) S_f^0(r+q), \quad (26)$$

with the orientation fixed separately for each diagram. These are alternative representations of the same hard photon-emission term.

5.4 Photon-DA matrix elements

The condensate-dependent photon terms are vacuum-to-photon nonlocal matrix elements. We use Rohrwild's background-field notation directly:

$$\begin{aligned} \langle 0 | \bar{s}(0) [0, x] \sigma_{\mu\nu} s(x) | 0 \rangle_F &= ee_s \langle \bar{s}s \rangle \int_0^1 du \left\{ \left[\chi_\gamma \phi_\gamma(u) + \frac{x^2}{16} \mathcal{A}(u) \right] F_{\mu\nu}(ux) \right. \\ &\quad \left. + \frac{\mathcal{B}(u)}{8} x^\rho [x_\nu F_{\mu\rho}(ux) - x_\mu F_{\nu\rho}(ux)] \right\}, \end{aligned} \quad (27)$$

and

$$\langle 0 | \bar{s}(0) [0, x] \gamma_\mu s(x) | 0 \rangle_F = -\frac{ee_s}{2} f_{3\gamma} \int_0^1 du \bar{\psi}^{(V)}(u) x^\rho F_{\rho\mu}(ux). \quad (28)$$

Thus no H_γ or $\bar{H}_\gamma$ is introduced in the Rohrwild formulae. The exact notation dictionary to the alternative h_γ representation is

$$\boxed{\mathcal{B}(u) = -4 \int_0^u d\alpha (u - \alpha) h_\gamma(\alpha)}, \quad \mathbb{A}(u) - 8\bar{H}_\gamma(u) = \mathcal{A}(u) + 2\mathcal{B}(u). \quad (29)$$

The factor two in the second equality is essential. It follows from the operator identity in the Ball–Braun–Kivel convention; replacing the Colangelo combination by $\mathcal{A} + \mathcal{B}$ would be incorrect. Similarly,

$$\bar{\psi}^{(V)}(u) = 2 \int_0^u dv \psi^v(v) = 2\Psi^v(u) \quad (30)$$

for the conformal models used here. The three-particle matrix elements

$$\langle \gamma | \bar{s}(x) g_s G_{\alpha\beta}(vx) \Gamma s(0) | 0 \rangle$$

define $\mathcal{S}, \tilde{\mathcal{S}}, \mathcal{T}_1, \dots, \mathcal{T}_4$. Their axial combination used in the sum rule is

$$\mathcal{F}_1 = \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4 + 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2). \quad (31)$$

6 Momentum routing

6.1 Heavy-line emission

The three fermion momenta are k , $k + q$, and $k - p$. The denominators are

$$d_{H1} = (k + q)^2 - m_Q^2, \quad (32)$$

$$d_{H2} = k^2 - m_Q^2, \quad (33)$$

$$d_{H3} = (k - p)^2 - m_s^2. \quad (34)$$

The current momenta are

$$k - (k - p) = p, \quad (k + q) - (k - p) = p + q. \quad (35)$$

6.2 Strange-line emission

The three momenta are k , $k - p$, and $k - p - q$. The denominators are

$$d_{S1} = k^2 - m_Q^2, \quad (36)$$

$$d_{S2} = (k - p)^2 - m_s^2, \quad (37)$$

$$d_{S3} = (k - p - q)^2 - m_s^2. \quad (38)$$

The corresponding current momenta are

$$k - (k - p) = p, \quad k - (k - p - q) = p + q. \quad (39)$$

This vertex check is important. The old trial route $k - p + q$ gives $k - (k - p + q) = p - q$ at the initial vertex and is therefore not the routing of Eq. (11). It is kept only in the dedicated routing audit as a diagnostic comparison.

7 Hard-emission Dirac traces

Charges, color, denominators, the QED vertex, propagator factors, and the global Wick sign are deliberately kept outside the numerator traces. For a generic basis vertex Γ_μ^X , the heavy-emission numerator is

$$N_{H,\mu\nu}^X = \text{Tr}_D \left[i\gamma_5 (\not{k} + m_Q) \gamma_\nu (\not{k} + \not{q} + m_Q) \Gamma_\mu^X (\not{k} - \not{p} + m_s) \right]. \quad (40)$$

The strange-emission numerator is

$$N_{s,\mu\nu}^X = \text{Tr}_D \left[i\gamma_5 (\not{k} + m_Q) \Gamma_\mu^X (\not{k} - \not{p} - \not{q} + m_s) \gamma_\nu (\not{k} - \not{p} + m_s) \right]. \quad (41)$$

The implementation is short enough to display in full:

```
traceReduce[expression_] :=
  Collect2[
    DiracSimplify[expression, DiracSubstitute67 -> True],
    {mQ, ms}
  ];

heavyTrace[current_] := traceReduce[DiracTrace[
  gammaP[] . propNumerator[k, mQ] . GA[nu] .
  propNumerator[k + q, mQ] . current[mu] .
  propNumerator[k - p, ms]
]];
```

```

strangeTrace[current_] := traceReduce[DiracTrace[
  gammaP[] . propNumerator[k, mQ] . current[mu] .
  propNumerator[k - p - q, ms] . GA[nu] .
  propNumerator[k - p, ms]
]];

nAHeavy = heavyTrace[gammaA];
nAStrange = strangeTrace[gammaA];
nBHeavy = heavyTrace[gammaB];
nBStrange = strangeTrace[gammaB];

```

Each of the four traces is recomputed after a cyclic rotation of the matrix chain. All four cyclic differences vanish exactly.

8 Fermion-line Ward identities

For equal masses on the photon-emitting segment and $B = A + q$,

$$(A + m) \not{q} (B + m) = D_B(A + m) - D_A(B + m), \quad (42)$$

where $D_A = A^2 - m^2$ and $D_B = B^2 - m^2$. Equation (42) shows explicitly that replacing the photon vertex by q_ν collapses one of the adjacent propagators, as required by the Ward–Takahashi identity.

The notebook checks Eq. (42) independently for the heavy line, with $A = k$ and $B = k + q$, and the strange line, with $A = l - q$ and $B = l$:

```

wardHeavyResidual = FCE[DiracSimplify[
  DiracGammaExpand[
    propNumerator[k, mQ] . GS[q] . propNumerator[k + q, mQ]
    - (SP[k + q, k + q] - mQ^2) propNumerator[k, mQ]
    + (SP[k, k] - mQ^2) propNumerator[k + q, mQ]
  ],
  DiracSubstitute67 -> True, DiracOrder -> True
]] // ScalarProductExpand // Simplify;

```

Both the heavy and strange residuals are zero.

9 E1 projection

The transverse E1 tensor is chosen as

$$S_{\mu\nu} = p_\nu q_\mu - (p \cdot q) g_{\mu\nu}. \quad (43)$$

For $q^2 = 0$,

$$q^\nu S_{\mu\nu} = 0. \quad (44)$$

The normalized projector is

$$\mathcal{P}_{E1}^{\mu\nu} = \frac{p^\nu q^\mu - (p \cdot q) g^{\mu\nu}}{2(p \cdot q)^2}, \quad (45)$$

and satisfies

$$\mathcal{P}_{E1}^{\mu\nu} S_{\mu\nu} = 1. \quad (46)$$

The code is

```

e1Tensor = FV[p, nu] FV[q, mu] - SP[p, q] MT[mu, nu];
e1Projector = e1Tensor/(2 SP[p, q]^2);

e1Project[expression_] := Collect2[
  ScalarProductExpand[
    Contract[expression e1Projector] /. kinematicRules
  ],
  {mQ, ms}
];

projAHeavy = e1Project[nAHeavy];
projAStrange = e1Project[nAStrange];
projBHeavy = e1Project[nBHeavy];
projBStrange = e1Project[nBStrange];

```

The projector normalization, photon Ward contraction, and contraction with a physical initial-state polarization are all checked symbolically.

10 Physical-state projection and the Colangelo limit

10.1 Rotation of the projected invariant amplitudes

Let $\hat{T}_A$ and $\hat{T}_B$ denote the continuum-subtracted, double-Borel E1 invariant amplitudes obtained from the two basis currents. Because the correlator is linear in the current at the initial-state vertex, the physical projections are

$$\begin{pmatrix} \hat{T}_1 \\ \hat{T}_2 \end{pmatrix} = R(\theta) \begin{pmatrix} \hat{T}_A \\ \hat{T}_B \end{pmatrix}, \quad R(\theta) = \begin{pmatrix} \sin \theta & \cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix}, \quad R^T R = \mathbf{1}. \quad (47)$$

The same rotation therefore applies after the E1 projection, after the OPE, and after the Borel transformation; these linear operations commute.

Define the pseudoscalar and physical axial-state residues by

$$\langle 0 | J_P | P(p) \rangle = \frac{f_P m_P^2}{m_Q + m_s}, \quad (48)$$

$$\langle 0 | J_i^\mu | H_i(p', \eta) \rangle = f_i M_i \eta^\mu, \quad i = 1, 2, \quad (49)$$

and use

$$\langle \gamma(q, \varepsilon) P(p) | H_i(p', \eta) \rangle = e G_i [(\varepsilon^* \cdot \eta)(p \cdot q) - (\varepsilon^* \cdot p)(\eta \cdot q)]. \quad (50)$$

The pole term in the projected, double-Borel invariant is then

$$\hat{T}_i^{\text{pole}} = \frac{f_P m_P^2}{m_Q + m_s} f_i M_i G_i \exp\left(-\frac{M_i^2}{\mathcal{M}_1^2} - \frac{m_P^2}{\mathcal{M}_2^2}\right). \quad (51)$$

Consequently the correctly normalized physical coupling is

$$G_i = \frac{m_Q + m_s}{f_P m_P^2 f_i M_i} \exp\left(\frac{M_i^2}{\mathcal{M}_1^2} + \frac{m_P^2}{\mathcal{M}_2^2}\right) \hat{T}_i^{\text{OPE}}. \quad (52)$$

If the basis residues for a given pole are defined with the same mass normalization, Eq. (52) becomes

$$G_1 = \frac{\sin \theta f_A^{(1)} G_A^{(1)} + \cos \theta f_B^{(1)} G_B^{(1)}}{f_1}, \quad (53)$$

$$G_2 = \frac{\cos \theta f_A^{(2)} G_A^{(2)} - \sin \theta f_B^{(2)} G_B^{(2)}}{f_2}. \quad (54)$$

Thus a bare sine–cosine rotation of already normalized couplings is valid only when the corresponding residues cancel.

10.2 Pole-side versus QCD-side kinematics

The distinction between the two sides of the sum rule is essential. Before the Borel transformation the QCD correlator depends on two independent virtualities:

$$q^2 = 0, \quad p' = p + q, \quad 2p \cdot q = p'^2 - p^2, \quad p^2 \text{ and } p'^2 \text{ independent.} \quad (55)$$

One must *not* replace these variables by hadron masses inside the theoretical numerator. The mass substitutions occur only in the residue of an isolated hadronic pole,

$$p^2|_{\text{pole}} = m_P^2, \quad p'^2|_{\text{pole}} = M_i^2, \quad p \cdot q|_{\text{pole}} = \frac{M_i^2 - m_P^2}{2}. \quad (56)$$

Likewise, u_0 is not a pre-Borel kinematic replacement. It is generated by the double Borel transform and becomes $1/2$ only after the equal-Borel choice. This corrects the former “physical-residue numerator prescription” used in the provisional Python tensor estimate.

10.3 Rohrwild decomposition before the Borel transformation

In the Rohrwild organization the two basis invariants have parallel decompositions,

$$T_A^{\text{RW}}(p'^2, p^2) = T_A^{\text{pert}} + T_A^{\text{tw}2} + T_A^{\text{tw}3} + T_A^{\text{tw}4} + T_A^{3\text{p},g} + T_A^{3\text{p},\gamma}, \quad (57)$$

$$T_B^{\text{RW}}(p'^2, p^2) = T_B^{\text{pert}} + T_B^{\text{tw}2} + T_B^{\text{tw}3} + T_B^{\text{tw}4} + T_B^{3\text{p},g} + T_B^{3\text{p},\gamma}. \quad (58)$$

There is no separate ordinary local-condensate term in either equation. Factors of $\langle \bar{s}s \rangle$ still occur in the normalization of *nonlocal* photon matrix elements such as $\chi_\gamma \langle \bar{s}s \rangle \phi_\gamma$; those factors must not be confused with insertion of the local propagator term $-\langle \bar{s}s \rangle/12$.

The perturbative contribution has the off-shell double-dispersion form

$$T_X^{\text{pert}}(p'^2, p^2) = \int ds_1 ds_2 \frac{\rho_X^{\text{pert}}(s_1, s_2)}{(s_1 - p'^2)(s_2 - p^2)}, \quad X = A, B. \quad (59)$$

For a two-particle photon DA define

$$D(u) = m_Q^2 - \bar{u} p'^2 - u p^2 = m_Q^2 - (p + \bar{u}q)^2, \quad \bar{u} = 1 - u. \quad (60)$$

Then every two-particle term can be kept explicitly off shell as

$$T_X^{2\text{p}} = \sum_{\Phi, n} \int_0^1 du \frac{C_{X, \Phi}^{(n)}(u, p^2, p \cdot q) \Phi(u)}{[D(u)]^n}, \quad \Phi \in \{\phi_\gamma, \bar{\psi}^{(V)}, \mathcal{A}, \mathcal{B}\}. \quad (61)$$

For a three-particle DA the analogous denominator is $D(\xi) = m_Q^2 - \xi p'^2 - (1 - \xi)p^2$, where ξ is the photon momentum fraction fixed by (α_i, v) , and

$$T_X^{3\text{p}} = \sum_{\mathcal{F}, n} \int \mathcal{D}\alpha_i \int_0^1 dv \frac{C_{X, \mathcal{F}}^{(n)}(\alpha_i, v, p^2, p \cdot q) \mathcal{F}(\alpha_i)}{[D(\xi)]^n}. \quad (62)$$

The current-dependent E1 trace kernels already obtained independently in *Mathematica* are listed below. Common charges, Wilson denominators, and DA normalizations are not included in the table. For the soft two-particle line, $k = p + \bar{u}q$ has been used, but no hadronic mass substitution has been made.

DA structure	A-current kernel	B-current kernel
two-particle tensor	8	$\frac{8m_Q}{m_Q + m_s}$
two-particle vector	$-\frac{6im_Q}{p \cdot q}$	$-\frac{2i}{m_Q + m_s} \frac{3p^2 + (2\bar{u} + 3)p \cdot q}{p \cdot q}$
$\mathcal{S}_\gamma, \sigma_{\alpha\beta}$ part	8	$\frac{8m_Q}{m_Q + m_s}$
$\mathcal{T}_4', \sigma_{\alpha\beta}$ part	-16i	$-\frac{16im_Q}{m_Q + m_s}$
$\mathcal{S}_\gamma, x_\alpha \gamma_\beta$ part	$-\frac{4i q \cdot x}{p \cdot q}$	0
$\mathcal{T}_4', x_\alpha \gamma_\beta$ part	0	0

Table 1: Explicit pre-Borel E1 kernels. These are QCD-side expressions and contain the independent invariants p^2 and $p \cdot q$ rather than hadron masses. The zeroes in the last column follow only after using the numerator-free $x_\alpha \mathcal{F}^{\alpha\beta} \gamma_\beta$ propagator term.

10.4 Exact double-Borel identities

Define

$$\frac{1}{M^2} = \frac{1}{\mathcal{M}_1^2} + \frac{1}{\mathcal{M}_2^2}, \quad u_0 = \frac{\mathcal{M}_1^2}{\mathcal{M}_1^2 + \mathcal{M}_2^2}, \quad A_n = \frac{(M^2)^{2-n}}{\Gamma(n)} \exp\left(-\frac{m_Q^2}{M^2}\right). \quad (63)$$

For $I_n[F] = \int_0^1 du F(u)/[D(u)]^n$, the double Borel transform is

$$\mathcal{B}_{\mathcal{M}_1^2}^{p'^2} \mathcal{B}_{\mathcal{M}_2^2}^{p^2} I_n[F] = A_n F(u_0). \quad (64)$$

For $n > 1$ and vanishing endpoint terms, numerator virtualities give

$$\mathcal{B}_{12} \int_0^1 du \frac{(p \cdot q) F(u)}{D(u)^n} = \frac{A_{n-1}}{2(n-1)} F'(u_0), \quad (65)$$

$$\mathcal{B}_{12} \int_0^1 du \frac{p'^2 F(u)}{D(u)^n} = m_Q^2 A_n F(u_0) - A_{n-1} F(u_0) + \frac{A_{n-1}}{n-1} [u F(u)]'_{u=u_0}, \quad (66)$$

$$\mathcal{B}_{12} \int_0^1 du \frac{p^2 F(u)}{D(u)^n} = m_Q^2 A_n F(u_0) - A_{n-1} F(u_0) - \frac{A_{n-1}}{n-1} [\bar{u} F(u)]'_{u=u_0}. \quad (67)$$

Here $\mathcal{B}_{12}$ abbreviates the two Borel operators in Eq. (64). If an endpoint does not vanish, its contact term must be retained. Equations (65)–(67) show explicitly why replacing p^2 and $p \cdot q$ by pole masses on the QCD side is not valid: the correct Borel images contain DA derivatives and possible endpoint terms.

The perturbative term transforms instead according to

$$\hat{T}_X^{\text{pert}} = \int_{\Omega(s_0)} ds_1 ds_2 \rho_X^{\text{pert}}(s_1, s_2) \exp\left(-\frac{s_1}{\mathcal{M}_1^2} - \frac{s_2}{\mathcal{M}_2^2}\right). \quad (68)$$

Thus a numerator p'^2 , p^2 , or $p \cdot q$ in a genuine double dispersion density maps to s_1 , s_2 , or $(s_1 - s_2)/2$, respectively, up to single-pole contact terms annihilated by the second Borel transform.

10.5 Explicit invariants after the Borel transformation

For the equal-Borel choice $\mathcal{M}_1^2 = \mathcal{M}_2^2 = 2M^2$, one has $u_0 = 1/2$. In the Rohrwild momentum routing used below it is convenient to write

$$a_0 = 1 - u_0 = \frac{\mathcal{M}_2^2}{\mathcal{M}_1^2 + \mathcal{M}_2^2}, \quad a_0 = \frac{1}{2} \quad \text{for equal Borel masses.} \quad (69)$$

The physical normalization factor and couplings are

$$\mathcal{N}_i(M^2) = \frac{\exp[(M_i^2 + m_P^2)/(2M^2)](m_Q + m_s)}{M_i f_i m_P^2 f_P}, \quad i = 1, 2, \quad (70)$$

$$G_1 = \mathcal{N}_1 \left(\sin \theta \hat{T}_A^{\text{RW}} + \cos \theta \hat{T}_B^{\text{RW}} \right), \quad (71)$$

$$G_2 = \mathcal{N}_2 \left(\cos \theta \hat{T}_A^{\text{RW}} - \sin \theta \hat{T}_B^{\text{RW}} \right). \quad (72)$$

The pole masses in $\mathcal{N}_i$ come from the phenomenological side and do not constitute a QCD-side substitution.

In Rohrwild notation the Borel invariants are

$$\hat{T}_A^{\text{RW}} = \hat{T}_A^{\text{pert}} + \hat{T}_A^{\text{tw}2} + \hat{T}_A^{\text{tw}3} + \hat{T}_A^{\text{tw}4} + \hat{T}_A^{3\text{p},g} + \hat{T}_A^{3\text{p},\gamma}, \quad (73)$$

$$\hat{T}_B^{\text{RW}} = \hat{T}_B^{\text{pert}} + \hat{T}_B^{\text{tw}2} + \hat{T}_B^{\text{tw}3} + \hat{T}_B^{\text{tw}4} + \hat{T}_B^{3\text{p},g} + \hat{T}_B^{3\text{p},\gamma}. \quad (74)$$

There is no $\hat{T}_X^{\langle \bar{s}s \rangle}$ slot.

The completed axial pieces not affected by the disputed on-shell shortcut are

$$\begin{aligned} \hat{T}_A^{\text{pert}} &= \int_{(m_Q+m_s)^2}^{s_0} ds e^{-s/M^2} \rho_A^P(s), \\ \hat{T}_A^{\text{tw}2} &= e_s \langle \bar{s}s \rangle (E_Q - E_0) M^2 \chi_\gamma \phi_\gamma(a_0), \\ \hat{T}_A^{\text{tw}3} &= e_s f_{3\gamma} m_Q E_Q \bar{\psi}^{(V)}(a_0), \\ \hat{T}_A^{\text{tw}4} &= \frac{e_s \langle \bar{s}s \rangle E_Q}{4} [\mathcal{A}(a_0) + 2\mathcal{B}(a_0)] \left(1 + \frac{m_Q^2}{M^2} \right), \\ \hat{T}_A^{3\text{p},g} &= e_s \langle \bar{s}s \rangle E_Q \mathcal{I}_{\mathcal{F}_1}(u_0), \end{aligned} \quad (75)$$

where

$$E_Q = e^{-m_Q^2/M^2}, \quad E_0 = e^{-s_0/M^2}, \quad (76)$$

The continuum prescription is term-specific. The leading twist-2 term carries $\Delta E_Q = E_Q - E_0$; twist-3, twist-4, and the gluonic three-particle contribution retain E_Q , as in the channel-specific $D_{s1} \rightarrow D_s \gamma$ sum rule. The separate gauge-completion electromagnetic three-particle term below follows Rohrwild's I_F -type ΔE_Q subtraction. The three-particle convolution is

$$\begin{aligned} \mathcal{I}_{\mathcal{F}_1}(u_0) &= \int_0^{1-u_0} dv \int_0^{u_0/(1-v)} d\alpha_g \mathcal{F}_1(u_0 - (1-v)\alpha_g, 1 - u_0 - v\alpha_g, \alpha_g) \\ &\quad + \int_{1-u_0}^1 dv \int_0^{(1-u_0)/v} d\alpha_g \mathcal{F}_1(u_0 - (1-v)\alpha_g, 1 - u_0 - v\alpha_g, \alpha_g). \end{aligned} \quad (77)$$

The condensate factors in these equations belong to nonlocal photon DAs.

The fully assembled gluonic and electromagnetic three-particle terms are given below in Eqs. (94) and (97). In particular, no unevaluated $\mathcal{B}_{12}[\dots]$ remains in the final expressions. The previously printed terms proportional to $m_P^2 - M_i^2$ applied Eq. (56) inside a QCD numerator and are withdrawn.

For the tensor current, the perturbative density used in the diagonal double-dispersion test is

$$\begin{aligned} \rho_B^P(s) &= e_s \mathcal{L}(s; m_s, m_Q; a_s, b_s) - e_Q \mathcal{L}(s; m_Q, m_s; a_Q, b_Q), \\ \mathcal{L}(s; m_i, m_j; a, b) &= -\frac{3}{8\pi^2} \left[2a \ln \frac{s - m_j^2 + m_i^2 - \sqrt{\lambda}}{s - m_j^2 + m_i^2 + \sqrt{\lambda}} + \frac{b}{m_i^2} \frac{m_j^2 - m_i^2 - s}{s^2} \sqrt{\lambda} \right], \end{aligned} \quad (78)$$

$$\begin{aligned}
d &= m_Q + m_s, & a_s &= \frac{m_Q m_s}{d}, & b_s &= \frac{m_s^2}{d}, \\
a_Q &= \frac{2m_Q^2 - m_Q m_s}{d}, & b_Q &= -\frac{m_Q^2 s}{d}.
\end{aligned} \tag{79}$$

The tensor-DA traces imply the exact, kinematics-independent relations

$$\hat{T}_B^{\text{tw}2} = \frac{m_Q}{d} \hat{T}_A^{\text{tw}2}, \quad \hat{T}_B^{\text{tw}4} = \frac{m_Q}{d} \hat{T}_A^{\text{tw}4}. \tag{80}$$

The vector DA contains p^2 and $p \cdot q$ before the transform. With

$$D_a = m_Q^2 - a p'^2 - (1-a)p^2, \tag{81}$$

its independently reduced off-shell kernels are

$$K_A^{(V)}(a) = -\frac{8m_Q}{D_a^2}, \quad K_B^{(V)}(a) = -\frac{8[m_Q^2 + (1-a)p \cdot q]}{dD_a^2}. \tag{82}$$

The exact $n = 2$ double-Borel maps in this variable are

$$\begin{aligned}
\mathcal{B}_{12} \int_0^1 da \frac{F(a)}{D_a^2} &= E_Q F(a_0), \\
\mathcal{B}_{12} \int_0^1 da \frac{(p \cdot q)F(a)}{D_a^2} &= -\frac{M^2 E_Q}{2} F'(a_0), \\
\mathcal{B}_{12} \int_0^1 da \frac{p^2 F(a)}{D_a^2} &= E_Q [m_Q^2 F(a_0) + M^2 a_0 F'(a_0)].
\end{aligned} \tag{83}$$

Consequently the requested explicit twist-3 tensor result is

$$\boxed{\hat{T}_B^{\text{tw}3} = \frac{e_s f_{3\gamma} E_Q}{d} \left[m_Q^2 \bar{\psi}^{(V)}(a_0) - \frac{M^2}{2} \frac{d}{da} \{ (1-a) \bar{\psi}^{(V)}(a) \} \Big|_{a=a_0} \right]}. \tag{84}$$

For the symmetric choice, $\bar{\psi}^{(V)}(1/2) = 0$, and hence

$$\hat{T}_A^{\text{tw}3} = 0, \quad \hat{T}_B^{\text{tw}3} = -\frac{e_s f_{3\gamma} M^2 E_Q}{4d} \bar{\psi}^{(V)'} \left(\frac{1}{2} \right). \tag{85}$$

Thus the vanishing of the axial vector-DA term at $a_0 = 1/2$ must not be copied to the tensor current.

Corrected post-Borel $x_\alpha \mathcal{F}^{\alpha\beta} \gamma_\beta$ kernels. For a two- or three-particle DA projected onto the heavy-line momentum fraction a , define its effective line density

$$F_{\text{eff}}(a) = \int \mathcal{D}\alpha_i \int_0^1 dv w_F(\alpha_i, v) \mathcal{F}(\alpha_i) \delta[a - a(\alpha_i, v)]. \tag{86}$$

The weight w_F contains the explicit v factor and the Lorentz weight of the DA. This propagator term has no $(\not{k} + m_Q)$ numerator. Writing $F_0 = F_{\text{eff}}(a_0)$ and $F_1 = F'_{\text{eff}}(a_0)$, the complete scalar Borel images of the Mathematica $x_\alpha \gamma_\beta$ E1 kernels are

$$\begin{aligned}
\mathfrak{X}_{A,S}[F] &= -8E_Q F_0, & \mathfrak{X}_{A,T_1}[F] &= 48iE_Q F_0, \\
\mathfrak{X}_{A,T_2}[F] &= 16iE_Q F_0, & \mathfrak{X}_{A,T_3}[F] &= -16iE_Q F_0, \\
\mathfrak{X}_{A,T_4}[F] &= 0,
\end{aligned} \tag{87}$$

$$\mathfrak{X}_{B,S}[F] = \mathfrak{X}_{B,T_1}[F] = \mathfrak{X}_{B,T_2}[F] = \mathfrak{X}_{B,T_3}[F] = \mathfrak{X}_{B,T_4}[F] = 0. \tag{88}$$

The formerly quoted nonzero B kernels resulted from multiplying this second propagator term by a free numerator that belongs only to the $\sigma_{\alpha\beta}$ term. The corresponding $\sigma_{\alpha\beta}$ pieces contain no numerator virtuality and are simply $E_Q F_{\text{eff}}(a_0)$ times the trace coefficients

$$\begin{array}{c|cc} & A & B \\ \hline \mathcal{S} \text{ or } \mathcal{S}_\gamma & 8 & 8m_Q/d \\ \mathcal{T}_4 \text{ or } \mathcal{T}_4^\gamma & -16i & -16im_Q/d \end{array}. \quad (89)$$

The following operator-level assembly supplies every Fierz, charge, DA and support weight rather than leaving them implicit.

Explicit operator line projections. For an operator-weighted DA $\mathcal{F}(\alpha_i, v)$, the fraction entering the heavy denominator is

$$a(\alpha_i, v) = \alpha_{\bar{q}} + v\alpha_g. \quad (90)$$

Define

$$\begin{aligned} \mathcal{J}_z[\mathcal{F}] &= \int_0^z dv \int_0^{(1-z)/(1-v)} d\alpha_g \mathcal{F}(1-z-(1-v)\alpha_g, z-v\alpha_g, \alpha_g, v) \\ &\quad + \int_z^1 dv \int_0^{z/v} d\alpha_g \mathcal{F}(1-z-(1-v)\alpha_g, z-v\alpha_g, \alpha_g, v). \end{aligned} \quad (91)$$

This is the explicit result of the $\delta[z - \alpha_{\bar{q}} - v\alpha_g]$ integration.

The $1/(q \cdot x)$ terms require one further support density. Performing Rohrwild's auxiliary- u delta integration before the Borel transformation gives

$$\begin{aligned} \mathcal{L}_z[X] &= \int_0^{1-z} d\alpha_q \int_0^{\alpha_q} d\alpha'_q \int_0^z d\alpha_{\bar{q}} \frac{z - \alpha_{\bar{q}}}{(1 - \alpha_q - \alpha_{\bar{q}})^2} X(\alpha_q, \alpha_{\bar{q}}, 1 - \alpha_q - \alpha_{\bar{q}}) \\ &\quad - \int_0^{1-z} d\alpha_q \int_z^{1-\alpha_q} d\beta \frac{z}{\beta^2} \int_0^\beta d\alpha'_{\bar{q}} X(\alpha_q, \alpha'_{\bar{q}}, 1 - \alpha_q - \alpha'_{\bar{q}}) \\ &\quad + \int_0^{1-z} d\alpha_q \frac{z}{(1 - \alpha_q)^2} \int_0^{\alpha_q} d\alpha'_q \int_0^{1-\alpha'_q} d\alpha_{\bar{q}} X(\alpha'_q, \alpha_{\bar{q}}, 1 - \alpha'_q - \alpha_{\bar{q}}). \end{aligned} \quad (92)$$

The lower limit $\beta = z$ in the second line is fixed by double-Borel support. Extending it to zero, as in the printed symmetric appendix, creates a spurious logarithmic divergence.

Complete gluonic contribution. All DAs in the following operator weights have arguments $(\alpha_q, \alpha_{\bar{q}}, \alpha_g)$:

$$\begin{aligned} \mathcal{F}_{A,\sigma}^g &= \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4, \\ \mathcal{F}_{A,x\gamma}^g &= 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2), \\ \mathcal{F}_A^g &= \mathcal{F}_{A,\sigma}^g + \mathcal{F}_{A,x\gamma}^g, \end{aligned} \quad \mathcal{F}_B^g = \frac{m_Q}{d} \mathcal{F}_{A,\sigma}^g. \quad (93)$$

Thus

$$\boxed{\hat{T}_A^{3p,g} = e_s \langle \bar{s}s \rangle E_Q \mathcal{J}_{a_0}[\mathcal{F}_A^g], \quad \hat{T}_B^{3p,g} = e_s \langle \bar{s}s \rangle E_Q \mathcal{J}_{a_0}[\mathcal{F}_B^g].} \quad (94)$$

In particular, $\mathcal{J}_{a_0}[\mathcal{F}_A^g] = \mathcal{I}_{\mathcal{F}_1}(u_0)$ for $a_0 = 1 - u_0$.

Complete electromagnetic contribution. The direct weights are

$$\mathcal{F}_A^\gamma = (1 - 2v)\mathcal{S}_\gamma - \mathcal{T}_4^\gamma, \quad \mathcal{F}_B^\gamma = \frac{m_Q}{d}(\mathcal{S}_\gamma - \mathcal{T}_4^\gamma). \quad (95)$$

The exact Borel treatment of the $\mathbb{P}$ numerator replaces the legacy pole-mass expression according to

$$\frac{2(m_P^2 - M_i^2)}{M^2} \mathcal{L}_{a_0}[\mathcal{T}_4^\gamma] \longrightarrow 2 \frac{d}{dz} \mathcal{L}_z[\mathcal{T}_4^\gamma] \Big|_{z=a_0}. \quad (96)$$

Writing $\Delta E_Q = E_Q - E_0$, the post-Borel invariants are

$$\begin{aligned} \boxed{\hat{T}_A^{3p,\gamma}} &= e_Q \langle \bar{s}s \rangle \Delta E_Q \left\{ \mathcal{J}_{a_0}[\mathcal{F}_A^\gamma] + 2 \frac{d}{dz} \mathcal{L}_z[\mathcal{T}_4^\gamma] \Big|_{z=a_0} \right\}, \\ \boxed{\hat{T}_B^{3p,\gamma}} &= e_Q \langle \bar{s}s \rangle \Delta E_Q \frac{m_Q}{d} \left\{ \mathcal{J}_{a_0}[\mathcal{S}_\gamma - \mathcal{T}_4^\gamma] + 2 \frac{d}{dz} \mathcal{L}_z[\mathcal{T}_4^\gamma] \Big|_{z=a_0} \right\}. \end{aligned} \quad (97)$$

Equations (91), (92), and (97) display every integration limit, operator weight and Borel derivative. No physical mass and no ordinary local condensate enters a QCD-side numerator.

For completeness, the axial perturbative density is

$$\begin{aligned} \rho_i(s; m_i, m_j, e_i) &= -\frac{3e_i}{8\pi^2} \left[2m_i \ln \frac{s - m_j^2 + m_i^2 - \lambda^{1/2}(s, m_j^2, m_i^2)}{s - m_j^2 + m_i^2 + \lambda^{1/2}(s, m_j^2, m_i^2)} \right. \\ &\quad \left. + (m_j - m_i) \frac{m_j^2 - m_i^2 - s}{s^2} \lambda^{1/2}(s, m_j^2, m_i^2) \right], \\ \rho_A^P(s) &= \rho_s(s; m_s, m_Q, e_s) - \rho_Q(s; m_Q, m_s, e_Q). \end{aligned} \quad (98)$$

10.6 Local-scheme separation

The alternative Colangelo axial comparison adds the ordinary local term

$$\hat{T}_A^{\text{Col,local}} = e_Q E_Q \langle \bar{s}s \rangle \left[1 - \frac{m_Q m_s}{M^2} + \frac{m_s^2}{2M^2} \left(1 + \frac{m_Q^2}{M^2} \right) \right]. \quad (99)$$

This term is absent from $\hat{T}_A^{\text{RW}}$ and $\hat{T}_B^{\text{RW}}$. It must never be added to the electromagnetic three-particle terms in Eq. (97).

Contribution	Rohrwild paper default	Colangelo comparison
Ordinary local condensate in transition LCSR	absent	retained for J_A
$\mathcal{S}_\gamma, \mathcal{T}_4^\gamma$	nonlocal; explicit post-Borel kernels	not added separately
Gluonic photon DAs	nonlocal	included through $\mathcal{F}_1$
Local condensates in two-point f_i sum rules	required	required
Local plus electromagnetic three-particle term	forbidden double counting	forbidden double counting

Table 2: Separation of the local-condensate prescriptions.

10.7 Exact pure-axial reduction to Colangelo et al.

Colangelo, De Fazio, and Ozpineci use the single axial current $J_\mu^A = \bar{Q} \gamma_\mu \gamma_5 s$ in their correlator. This is the Hermitian-conjugate orientation of our A current and has the same invariant

amplitude. In the convention of Eq. (47), their $D_{s1}(2460)$ calculation is obtained from the low channel by choosing

$$\boxed{\theta = \frac{\pi}{2} = 90^\circ}, \quad J_1 = J_A, \quad J_2 = -J_B. \quad (100)$$

At this point

$$\hat{T}_1 = \hat{T}_A, \quad f_1 = f_A, \quad G_1 = G_A. \quad (101)$$

For the equal-Borel convention $\mathcal{M}_1^2 = \mathcal{M}_2^2 = 2M^2$, Eq. (52) reduces to

$$G_1|_{\theta=\pi/2} = \frac{\exp[(M_1^2 + m_P^2)/(2M^2)](m_Q + m_s)}{M_1 f_A m_P^2 f_P} \hat{T}_A^{\text{OPE}}. \quad (102)$$

For $Q = c$, $P = D_s$, and $M_1 = m_{D_{s1}}$, the right-hand side is exactly the prefactor and axial OPE appearing in Eq. (4.4) of [Colangelo, De Fazio, and Ozpineci](#). Therefore the relation is analytic, not merely a numerical similarity. Choosing $\theta = 0$ instead places the pure axial current in the second row of Eq. (47); it swaps the low/high labels and is not our $D_{s1}(2460)$ convention.

The numerical pure- A comparison is shown in Table 3.

Calculation	G [GeV $^{-1}$]	Γ [keV]
Colangelo et al., single J_A	−0.37 to −0.29	19 to 29
This project, Stage 2 at $\theta = 90^\circ$	−0.3978 to −0.3124	20.51 to 33.26
Interval overlap	−0.3700 to −0.3124	20.51 to 29.00

Table 3: Pure-axial benchmark. The remaining displacement is caused by the input set and convention choices, not by the state projector.

No intermediate angle is fitted here. Formally, at fixed residues, $G_1(\theta) = a \sin \theta + b \cos \theta$ with $a = f_A^{(1)} G_A^{(1)} / f_1$ and $b = f_B^{(1)} G_B^{(1)} / f_1$. A physical angle extraction requires the complete tensor-current transition amplitude. The angle-dependent physical residue $f_1(\theta)$ is now supplied independently by the explicit $AA/AB/BB$ two-point matrix below.

10.8 Decay constants from the rotated two-point correlator

Introduce the transverse two-point matrix

$$\Pi^{\text{basis}} = \begin{pmatrix} \Pi_{AA} & \Pi_{AB} \\ \Pi_{AB} & \Pi_{BB} \end{pmatrix}. \quad (103)$$

The physical matrix is $\Pi^{\text{phys}} = R \Pi^{\text{basis}} R^T$, so that

$$\Pi_{11} = \sin^2 \theta \Pi_{AA} + \cos^2 \theta \Pi_{BB} + 2 \sin \theta \cos \theta \Pi_{AB}, \quad (104)$$

$$\Pi_{22} = \cos^2 \theta \Pi_{AA} + \sin^2 \theta \Pi_{BB} - 2 \sin \theta \cos \theta \Pi_{AB}, \quad (105)$$

$$\Pi_{12} = \sin \theta \cos \theta (\Pi_{AA} - \Pi_{BB}) + (\cos^2 \theta - \sin^2 \theta) \Pi_{AB}. \quad (106)$$

This is already the useful paper-level explicit form. If each basis correlator is decomposed as

$$\hat{\Pi}_{XY} = \int_{s_{\text{th}}}^{s_0} ds e^{-s/\mathcal{M}^2} \rho_{XY}^{\text{pert}}(s) + \sum_d \hat{\Pi}_{XY}^{(d)}, \quad XY = AA, BB, AB, \quad (107)$$

All three perturbative densities are explicit. With

$$d = m_Q + m_s, \quad \lambda_2(s) = [s - (m_Q + m_s)^2][s - (m_Q - m_s)^2], \quad (108)$$

they are

$$\boxed{\rho_{AA}(s)} = \frac{\sqrt{\lambda_2(s)}}{8\pi^2} \left[2 - \frac{m_Q^2 + m_s^2 + 6m_Q m_s}{s} - \frac{(m_Q^2 - m_s^2)^2}{s^2} \right], \quad (109)$$

$$\boxed{\rho_{AB}(s) = \rho_{BA}(s)} = \frac{\sqrt{\lambda_2(s)}}{8\pi^2} \frac{3(m_s - m_Q)(d^2 - s)}{ds}, \quad (110)$$

$$\boxed{\rho_{BB}(s)} = \frac{\sqrt{\lambda_2(s)}}{8\pi^2} \frac{-(d^2 - s)(2m_s^2 - 4m_Q m_s + 2m_Q^2 + s)}{d^2 s}. \quad (111)$$

The AA expression is algebraically identical to the Colangelo axial-current density, and $\rho_{AB} = \rho_{BA}$ is an exact trace check.

At the common $\text{LO}+d = 3+d = 5$ truncation, the remaining Borel matrix is also explicit:

$$\hat{\Pi}(\mathcal{M}^2, s_0) = \int_{d^2}^{s_0} ds e^{-s/\mathcal{M}^2} \begin{pmatrix} \rho_{AA} & \rho_{AB} \\ \rho_{AB} & \rho_{BB} \end{pmatrix} + e^{-m_Q^2/\mathcal{M}^2} [C_3(\mathcal{Q}_0 + \mathcal{Q}_1 + \mathcal{Q}_2) + C_5 \mathcal{Q}_5], \quad (112)$$

where $C_3 = \langle \bar{s}s \rangle$, $C_5 = \langle \bar{s}g_s \sigma \cdot G s \rangle$, and

$$\begin{aligned} \mathcal{Q}_0 &= \begin{pmatrix} m_Q & m_Q^2/d \\ m_Q^2/d & m_Q^3/d^2 \end{pmatrix}, \\ \mathcal{Q}_1 &= \begin{pmatrix} -\frac{m_s m_Q^2}{2\mathcal{M}^2} & \frac{m_s m_Q(1 - m_Q^2/\mathcal{M}^2)}{2d} \\ \frac{m_s m_Q(1 - m_Q^2/\mathcal{M}^2)}{2d} & \frac{m_s m_Q^2(1 - m_Q^2/(2\mathcal{M}^2))}{d^2} \end{pmatrix}, \\ \mathcal{Q}_2 &= \begin{pmatrix} \frac{m_s^2 m_Q^3}{m_s^2(m_Q^4/\mathcal{M}^4 - m_Q^2/\mathcal{M}^2 - 1)} & \frac{m_s^2(m_Q^4/\mathcal{M}^4 - m_Q^2/\mathcal{M}^2 - 1)}{m_s^2(m_Q^5/(2\mathcal{M}^4) - m_Q^3/\mathcal{M}^2)} \\ \frac{m_s^2(m_Q^4/\mathcal{M}^4 - m_Q^2/\mathcal{M}^2 - 1)}{m_s^2(m_Q^5/(2\mathcal{M}^4) - m_Q^3/\mathcal{M}^2)} & \frac{m_s^2(m_Q^5/(2\mathcal{M}^4) - m_Q^3/\mathcal{M}^2)}{d^2} \end{pmatrix}, \\ \mathcal{Q}_5 &= \begin{pmatrix} -\frac{m_Q^3}{4\mathcal{M}^4 - m_Q^2/\mathcal{M}^2 - 1} & -\frac{m_Q^4/\mathcal{M}^4 - m_Q^2/\mathcal{M}^2 - 1}{m_Q^5/(4\mathcal{M}^4) - m_Q^3/(2\mathcal{M}^2)} \\ -\frac{m_Q^4/\mathcal{M}^4 - m_Q^2/\mathcal{M}^2 - 1}{m_Q^5/(4\mathcal{M}^4) - m_Q^3/(2\mathcal{M}^2)} & -\frac{m_Q^5/(4\mathcal{M}^4) - m_Q^3/(2\mathcal{M}^2)}{d^2} \end{pmatrix}. \end{aligned} \quad (113)$$

For compactness write $s_\theta = \sin \theta$, $c_\theta = \cos \theta$ and define the scalar projections of each displayed local matrix by

$$\begin{aligned} q_1^{(n)} &= s_\theta^2(\mathcal{Q}_n)_{AA} + c_\theta^2(\mathcal{Q}_n)_{BB} + 2s_\theta c_\theta(\mathcal{Q}_n)_{AB}, \\ q_2^{(n)} &= c_\theta^2(\mathcal{Q}_n)_{AA} + s_\theta^2(\mathcal{Q}_n)_{BB} - 2s_\theta c_\theta(\mathcal{Q}_n)_{AB}, \quad n = 0, 1, 2, 5. \end{aligned} \quad (114)$$

Because every entry of $\mathcal{Q}_n$ is given in Eq. (113), Eq. (114) is a fully explicit scalar expression, not a new phenomenological input.

The two poles use separate continuum thresholds $s_0^{(1)}$ and $s_0^{(2)}$, fitted to M_1 and M_2 , respectively. The continuum-subtracted quantities actually evaluated are

$$\begin{aligned} \hat{\Pi}_{11} &= \int_{d^2}^{s_0^{(1)}} ds e^{-s/\mathcal{M}^2} \left[s_\theta^2 \rho_{AA}(s) + c_\theta^2 \rho_{BB}(s) + 2s_\theta c_\theta \rho_{AB}(s) \right] \\ &\quad + e^{-m_Q^2/\mathcal{M}^2} \left\{ C_3[q_1^{(0)} + q_1^{(1)} + q_1^{(2)}] + C_5 q_1^{(5)} \right\}, \end{aligned} \quad (115)$$

$$\begin{aligned} \hat{\Pi}_{22} &= \int_{d^2}^{s_0^{(2)}} ds e^{-s/\mathcal{M}^2} \left[c_\theta^2 \rho_{AA}(s) + s_\theta^2 \rho_{BB}(s) - 2s_\theta c_\theta \rho_{AB}(s) \right] \\ &\quad + e^{-m_Q^2/\mathcal{M}^2} \left\{ C_3[q_2^{(0)} + q_2^{(1)} + q_2^{(2)}] + C_5 q_2^{(5)} \right\}. \end{aligned} \quad (116)$$

The final square-root formulas used in the numerical code are therefore

$$\boxed{f_1(\mathcal{M}^2, s_0^{(1)}, \theta)} = \frac{e^{M_1^2/(2\mathcal{M}^2)}}{M_1} \sqrt{\widehat{\Pi}_{11}(\mathcal{M}^2, s_0^{(1)}, \theta)}, \quad (117)$$

$$\boxed{f_2(\mathcal{M}^2, s_0^{(2)}, \theta)} = \frac{e^{M_2^2/(2\mathcal{M}^2)}}{M_2} \sqrt{\widehat{\Pi}_{22}(\mathcal{M}^2, s_0^{(2)}, \theta)}. \quad (118)$$

The condition $\widehat{\Pi}_{12} \simeq 0$ can be used to diagnose or determine the preferred mixing angle. At $\theta = \pi/2$, Eq. (117) becomes the pure axial-current result

$$f_1 = \frac{e^{M_1^2/(2\mathcal{M}^2)}}{M_1} \sqrt{\widehat{\Pi}_{AA}}, \quad (119)$$

which is the decay-constant normalization used by Colangelo et al.

10.9 Current two-point result for f_1 and f_2

Equations (109)–(118) replace the former external- f_1 /overlap normalization. At the fixed angle from our previous study,

$$\theta_{D_s} = 26.6 \pm 0.6^\circ, \quad (120)$$

the fixed-angle, two-threshold Monte Carlo gives

$$f_1 = 0.4046[0.3799, 0.4291] \text{ GeV}, \quad f_2 = 0.1677[0.1594, 0.1756] \text{ GeV}. \quad (121)$$

For example, the explicit central Mathematica substitution

$$\mathcal{M}^2 = 2.35 \text{ GeV}^2, \quad s_0^{(1)} = 7.925 \text{ GeV}^2, \quad s_0^{(2)} = 9.975 \text{ GeV}^2, \quad \theta = 26.6^\circ \quad (122)$$

with $m_c = 1.27 \text{ GeV}$, $m_s = 0.093 \text{ GeV}$, $\langle \bar{s}s \rangle = 0.8[-(0.24 \text{ GeV})^3]$ and $m_0^2 = 0.8 \text{ GeV}^2$ gives

$$f_1 = 0.4005893 \text{ GeV}, \quad f_2 = 0.1666704 \text{ GeV}. \quad (123)$$

The quoted values in Eq. (121) are the medians and 16–84 percentile intervals after varying the QCD inputs, Borel parameter, threshold fits and the external angle. No decay constant is externally anchored in this result. Diagonalizing the same truncated matrix gives $\theta_{\text{matrix}} = 30.63[29.94, 31.29]^\circ$; this is retained as an OPE-truncation diagnostic rather than replacing the dedicated external angle. The Python and Mathematica fixed-point residues agree below $3 \times 10^{-8} \text{ GeV}$. Local condensates are required in this ordinary two-point sum rule; their presence here does not reintroduce a local condensate into the external-photon transition LCSR.

Finally, once G_i is normalized with the corresponding f_i , the decay width is

$$\Gamma(H_i \rightarrow P\gamma) = \frac{\alpha_{\text{em}}}{3} |G_i|^2 k_{\gamma,i}^3, \quad k_{\gamma,i} = \frac{M_i^2 - m_P^2}{2M_i}. \quad (124)$$

11 Inverse-denominator reduction

The projected expressions contain k^2 , $k \cdot p$, and $k \cdot q$. For the heavy-emission topology these are replaced by

$$k^2 = d_{H2} + m_Q^2, \quad (125)$$

$$k \cdot q = \frac{d_{H1} - d_{H2}}{2}, \quad (126)$$

$$k \cdot p = \frac{d_{H2} + m_Q^2 + p^2 - m_s^2 - d_{H3}}{2}. \quad (127)$$

For strange emission,

$$k^2 = d_{S1} + m_Q^2, \quad (128)$$

$$k \cdot p = \frac{d_{S1} + m_Q^2 + p^2 - m_s^2 - d_{S2}}{2}, \quad (129)$$

$$k \cdot q = p \cdot q + \frac{d_{S2} - d_{S3}}{2}. \quad (130)$$

Setting all three inverse denominators of a topology to zero extracts the coefficient of the genuine three-denominator triangle. Terms containing one or more d_i cancel propagators and are retained separately in the full machine output.

The axial triangle cores are

$$N_H^{A,\text{core}} = 4im_Q + \frac{4im_Q^3 - 4im_Q^2 m_s}{p \cdot q}, \quad (131)$$

$$N_s^{A,\text{core}} = -4im_s + \frac{4im_Q m_s^2 - 4im_s^3}{p \cdot q}. \quad (132)$$

The tensor-current cores are

$$N_H^{B,\text{core}} = \frac{8im_Q^2 - 4im_Q m_s}{m_Q + m_s} + \frac{4im_Q^2 p^2}{(m_Q + m_s)(p \cdot q)}, \quad (133)$$

$$N_s^{B,\text{core}} = \frac{-4im_Q m_s + 8im_s^2}{m_Q + m_s} + \frac{4im_s^2 p^2}{(m_Q + m_s)(p \cdot q)}. \quad (134)$$

These expressions refer only to the numerator traces of Eqs. (40)–(41); the charge, color, propagator, QED-vertex, and global Wick factors must still be attached.

12 Feynman parameterization of the hard triangles

For three denominators,

$$\frac{1}{D_1 D_2 D_3} = 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xD_1 + yD_2 + (1-x-y)D_3]^3}. \quad (135)$$

Equivalently one may use $x + y + z = 1$.

12.1 Heavy emission

Assign x, y, z to d_{H1}, d_{H2}, d_{H3} . Direct completion of the square gives

$$xd_{H1} + yd_{H2} + zd_{H3} = (k + xq - zp)^2 - \Delta_H, \quad (136)$$

where

$$\Delta_H = (x + y)m_Q^2 + zm_s^2 - xzp'^2 - yzp^2. \quad (137)$$

12.2 Strange emission

Assign x, y, z to d_{S1}, d_{S2}, d_{S3} . Then

$$xd_{S1} + yd_{S2} + zd_{S3} = [k - yp - z(p + q)]^2 - \Delta_s, \quad (138)$$

with

$$\Delta_s = xm_Q^2 + (y + z)m_s^2 - xyp^2 - xzp'^2. \quad (139)$$

The independent pairwise-invariant formula

$$\Delta = \sum_i x_i m_i^2 - \sum_{i < j} x_i x_j (r_i - r_j)^2 \quad (140)$$

reproduces Eqs. (137) and (139). The notebook checks both the completed-square and pairwise forms:

```
deltaH = (x + y) mQ^2 + z ms^2 - x z pPrime2 - y z p2;
shiftH = x q - z p;
heavyResidual = simplexReduce[
  x dH1 + y dH2 + z dH3
  - (SP[k + shiftH, k + shiftH] - deltaH)
];

deltaS = x mQ^2 + (y + z) ms^2 - x y p2 - x z pPrime2;
shiftS = -y p - z (p + q);
strangeResidual = simplexReduce[
  x dS1 + y dS2 + z dS3
  - (SP[k + shiftS, k + shiftS] - deltaS)
];
```

Both residuals vanish exactly.

13 Soft two-particle Fierz traces

After contracting the heavy quark and Fierz rearranging the strange bilocal, the Dirac factor is

$$T_i^X = \text{Tr}_D[\Gamma_P(\not{k} + m_Q)\Gamma_\mu^X\Gamma_i]. \quad (141)$$

The notebook evaluates the standard scalar, pseudoscalar, vector, axial, and tensor basis. The nonzero results for the axial current are

$$T_1^A = 4ik_\mu, \quad (142)$$

$$T_{\gamma_\rho}^A = -4im_Q g_{\mu\rho}, \quad (143)$$

$$T_{\sigma_{\rho\lambda}}^A = 4g_{\lambda\mu}k_\rho - 4g_{\mu\rho}k_\lambda. \quad (144)$$

For the tensor current,

$$T_{\gamma_\rho}^B = \frac{4i}{m_Q + m_s}[k_\mu p'_\rho - g_{\mu\rho}(k \cdot p')], \quad (145)$$

$$T_{\gamma_\rho\gamma_5}^B = -\frac{4}{m_Q + m_s}\epsilon_{\mu\rho k p'}, \quad (146)$$

$$T_{\sigma_{\rho\lambda}}^B = \frac{4m_Q}{m_Q + m_s}[g_{\lambda\mu}p'_\rho - g_{\mu\rho}p'_\lambda]. \quad (147)$$

All omitted entries vanish. Cyclically rotating the matrix chain gives the same tensor- and vector-basis traces for both currents.

14 Implemented OPE terms: exact status

The Mathematica restart now contains the complete analytic operator-weighted post-Borel assembly. The Python and Mathematica numerical layers now use the same corrected off-shell expressions; the older pole-mass numerator routines remain only as explicitly labeled historical diagnostics. The present status is:

OPE contribution	Axial current A	Tensor current B
Perturbative photon from Q and s	Published spectral density implemented numerically; corrected traces independently reproduced	Corrected traces and triangle cores derived; diagonal double-dispersion spectral mapping implemented independently in Python and Mathematica
Local $e_Q \langle \bar{s}s \rangle$ term	Excluded in the Rohrwild result; retained only in the Colangelo comparison with m_s, m_s^2 corrections	Excluded in the Rohrwild result; no Colangelo J_B term invented
Electromagnetic three-particle DAs, $\mathcal{S}_\gamma, \mathcal{T}_4^\gamma$	Complete support-corrected operator convolution is explicit	Complete support-corrected operator convolution is explicit; the former pole-mass numerator substitution is withdrawn
Twist-2 two-particle DA, $\chi \langle \bar{s}s \rangle \phi_\gamma$	Included in the exact numerical invariant	Exact factor $m_Q/(m_Q + m_s)$ included
Twist-4 two-particle DAs, $\mathcal{A}$ and $\mathcal{B}$	Explicit in the Rohrwild combination $\mathcal{A} + 2\mathcal{B}$	Exact factor $m_Q/(m_Q + m_s)$ relative to the axial piece
Twist-3 two-particle DA, $f_{3\gamma} \bar{\psi}^{(V)}$	Explicit; vanishes at the symmetric Borel point	Exact post-Borel derivative expression in Eq. (84)
Three-particle photon DAs, $\mathcal{S}, \tilde{\mathcal{S}}, \mathcal{T}_{1\dots 4}$	Full operator-weighted $\mathcal{F}_1$ line projection is explicit	Full operator-weighted line projection is explicit; the $x_\alpha \gamma_\beta$ trace vanishes
Mixed condensate $\langle \bar{s}g_s \sigma \cdot Gs \rangle$	Not included in the radiative three-point sum rule	Not included
Vacuum gluon condensate $\langle G^2 \rangle$	Not included in the radiative three-point sum rule	Not included

Table 4: Term-by-term status of the completed analytic and numerical transition calculation.

14.1 Completed numerical checkpoint

At $a_0 = 1/2$ the independently evaluated DA convolutions are

$$\begin{aligned}
\mathcal{J}[\mathcal{F}_{A,\sigma}^g] &= -0.1562500000, & \mathcal{J}[\mathcal{F}_{A,x\gamma}^g] &= +0.2265798500, \\
\mathcal{J}[\mathcal{F}_A^g] &= +0.0703298500, & \mathcal{J}[\mathcal{F}_A^\gamma] &= 0, \\
\mathcal{J}[\mathcal{S}_\gamma - \mathcal{T}_4^\gamma] &= 1.5625000000, & \mathcal{L}[\mathcal{T}_4^\gamma] &= 0.1111834583, \\
\mathcal{L}'[\mathcal{T}_4^\gamma] &= 0.4098669166.
\end{aligned} \tag{148}$$

For the charm reference point $M^2 = 3.75 \text{ GeV}^2$, $s_0 = 8.0 \text{ GeV}^2$ and the lattice photon normalization, the totals are

$$\hat{T}_A = -0.0311671290 \text{ GeV}^3, \quad \hat{T}_B = -0.0701320395 \text{ GeV}^3. \tag{149}$$

With $\theta = 26.6^\circ$ and the central projected residues this gives

$$\begin{aligned}
G_1 &= -0.4113423853 \text{ GeV}^{-1}, & \Gamma_1 &= 35.5665446 \text{ keV}, \\
G_2 &= +0.0464986762 \text{ GeV}^{-1}, & \Gamma_2 &= 0.6709364 \text{ keV}.
\end{aligned} \tag{150}$$

These are single-point regression values rather than final-window medians. For the direct bottom two-point reference $M_{2\text{pt}}^2 = 7.00 \text{ GeV}^2$, $s_0^{(1)} = 44.35 \text{ GeV}^2$, and $s_0^{(2)} = 43.025 \text{ GeV}^2$, the independent Mathematica evaluation gives

$$f_1^{(B_s)} = 0.5381107 \text{ GeV}, \quad f_2^{(B_s)} = 0.0888787 \text{ GeV}, \quad (151)$$

in agreement with Python. The direct B_s Monte Carlo gives $f_1^{(B_s)} = 0.5358[0.5096, 0.5586] \text{ GeV}$ and $f_2^{(B_s)} = 0.08896[0.08496, 0.09278] \text{ GeV}$. The cross-code transition comparison contains 145 common numerical entries and passes the 10^{-8} absolute-or-relative tolerance. The full fixed-seed uncertainty scan gives

$$\begin{aligned} \Gamma[D_{s1}(2460) \rightarrow D_s \gamma] &= 34.96[28.06, 43.53] \text{ keV}, \\ \Gamma[D_{s1}(2536) \rightarrow D_s \gamma] &= 0.988[0.092, 3.555] \text{ keV}, \\ \Gamma[B_{s1}(5750) \rightarrow B_s \gamma] &= 11.34[9.144, 14.06] \text{ keV}, \\ \Gamma[B_{s1}(5830) \rightarrow B_s \gamma] &= 8.702[6.474, 11.23] \text{ keV}. \end{aligned} \quad (152)$$

15 Verification summary and present boundary

Check	Result	Exact tests
Currents, Wick signs, traces, E1 cores, soft trace cyclicity	PASS	19/19
Standard vertex routing	PASS	4/4 momenta
Heavy- and strange-line Ward identities	PASS	2/2
Shifted and pairwise Feynman denominators	PASS	4/4
State rotation, OPE assembly, Colangelo limit, decay constant, and width	PASS	15/15
Notebook construction and corrected numerical section	PASS	53 cells; block run
Python–Mathematica transition regression	PASS	145/145 quantities
Direct B_s two-point central regression	PASS	f_1, f_2 , thresholds, pole/OPE diagnostics

Table 5: Status of the independent Mathematica/FeynCalc restart.

The B_s channel uses the same symbolic expressions with $m_Q \rightarrow m_b$ and the appropriate heavy-quark charge. Its direct $AA/AB/BB$ two-point normalization is implemented independently in `scripts/mathematica_twopoint_bs1_matrix_check.wl`; no Pullin–Zwicky decay constant or overlap closure enters the final result.

The completed symbolic and numerical stages now include the exact two-particle vector-DA Borel reduction, the corrected $x_\alpha \gamma_\beta$ kernels, the full operator-weighted gluonic and electromagnetic line projections, and the physical-state uncertainty scan. Useful higher-order follow-up tasks are:

1. independently rederive the shifted hard-loop double discontinuity directly from Eqs. (137)–(139), as a check beyond the present published spectral-density implementation;
2. add perturbative α_s corrections consistently to the transition and two-point sectors;
3. extend the common OPE truncation to the omitted higher-dimensional terms.

Python results that use the former physical-residue numerator shortcut remain historical diagnostics and are not used in the quoted tables.

A Reproduction commands

From papers/Ds_Bs_gamma/, run

```
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/mathematica_current_wick_trace_audit.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/mathematica_correlator_routing_audit.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/mathematica_hard_loop_parameterization.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/mathematica_state_mixing_projection.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/step13_soft_vector_da_borel.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/step14_explicit_double_borel_forms.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/step15_complete_three_particle_borel.wl  
  
python3 scripts/corrected_transition_central_python.py  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noinit -noprompt \  
-script scripts/mathematica_corrected_transition_numerics.wl  
  
python3 scripts/compare_corrected_transition_numerics.py  
  
python3 scripts/final_window_monte_carlo.py
```

To regenerate the notebook from its source cells, run

```
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noprompt \  
-script scripts/build_dsbs_symbolic_notebook.wl  
  
/Applications/Wolfram.app/Contents/MacOS/WolframKernel -noinit -noprompt \  
-script scripts/check_dsbs_symbolic_notebook.wl
```